Current Electricity: Revision Notes

Class 12 Physics level. Original sample notes written for testing the VivaVoice project.

1. Electric Current

Electric current is the rate of flow of electric charge through a cross-section of a conductor. If a net charge q passes through a cross-section in time t, the average current is:

$$I = q / t$$

The SI unit of current is the ampere (A). One ampere is a flow of one coulomb of charge per second. Current is a scalar quantity, even though we show its direction with an arrow, because currents add algebraically and do not follow the laws of vector addition.

By convention, the direction of current is the direction in which positive charges would move. In metals the charge carriers are free electrons, so the electrons actually move opposite to the conventional current.

Drift velocity

Free electrons in a metal move randomly at high speeds, but with no field their average velocity is zero. When an electric field E is applied, the electrons gain a small average velocity opposite to the field, called the drift velocity:

$$v_d = (e E \tau) / m$$

Here e is the electron charge, m its mass, and τ the relaxation time: the average time between successive collisions. Drift velocity is very small, of the order of 10^{-4} m/s, yet a bulb lights almost instantly because the electric field travels through the wire at nearly the speed of light and sets all electrons drifting at once.

Current and drift velocity

For a conductor with n free electrons per unit volume and cross-sectional area A:

$$I = n e A v_d$$

Current density j is current per unit area, $j = I / A = n e v_d$. Its unit is A/m^2, and it is a vector.

2. Ohm's Law and Resistance

Ohm's law states that the current through a conductor is directly proportional to the potential difference across its ends, provided physical conditions such as temperature stay constant:

$$V = I R$$

The constant R is the resistance of the conductor. Its SI unit is the ohm (Ω). One ohm is the resistance of a conductor through which one ampere flows when one volt is applied across it.

Resistivity

Resistance depends on the length L and cross-sectional area A of the conductor and on its material:

$$R = \rho L / A$$

ρ (rho) is the resistivity, a property of the material alone, measured in Ω m. Doubling the length doubles the resistance; doubling the area halves it. From the microscopic picture, $\rho = m / (n e^2 \tau)$.

Conductivity σ is the reciprocal of resistivity, $\sigma = 1/\rho$. Ohm's law in vector form is $j = \sigma E$.

Ohmic and non-ohmic devices

A device is ohmic if its V-I graph is a straight line through the origin, for example a metal wire at constant temperature. Non-ohmic devices break this in different ways:

- The V-I graph is not a straight line (for example, a filament bulb as it heats up).
- The current depends on the direction of V (for example, a diode).
- More than one value of V gives the same current (for example, gallium arsenide).

Temperature dependence

For metals, resistivity rises roughly linearly with temperature:

$$\rho_T = \rho_0 [1 + \alpha (T - T_0)]$$

α is the temperature coefficient of resistivity. It is positive for metals because hotter ions vibrate more, electrons collide more often, and τ decreases. Alloys such as nichrome and manganin have a very small α, which is why they are used in standard resistors and heating elements. Semiconductors have a negative α: heating frees more charge carriers, so n increases and resistivity falls.

3. Combination of Resistors

Series combination

Resistors in series carry the same current, and the total potential difference is the sum of the individual drops:

$$R_s = R_1 + R_2 + R_3 + \dots$$

The equivalent resistance in series is always greater than the largest individual resistance.

Parallel combination

Resistors in parallel have the same potential difference across them, and the total current divides among them:

$$1 / R_p = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots$$

The equivalent resistance in parallel is always less than the smallest individual resistance. For two resistors, $R_p = R_1 R_2 / (R_1 + R_2)$. Household appliances are connected in parallel so each gets the full mains voltage and can be switched on or off independently.

Worked example

Three resistors of 2 Ω, 3 Ω and 6 Ω are connected in parallel. $1/R_p = 1/2 + 1/3 + 1/6 = 1$, so $R_p = 1$ Ω. In series, the same three would give 11 Ω.

4. EMF and Internal Resistance

The electromotive force (EMF, ε) of a cell is the potential difference between its terminals when no current is drawn. Despite its name, it is not a force; it is the work done by the cell per unit charge, measured in volts.

Real cells have an internal resistance r. When the cell drives a current I through an external resistance R:

$$I = \varepsilon / (R + r) \text{ and } V = \varepsilon - I r$$

So the terminal voltage V is less than the EMF while the cell supplies current. While the cell is being charged, $V = \varepsilon + I r$, which is greater than the EMF.

Cells in series and parallel

- n identical cells in series: total EMF nε, total internal resistance nr.
- n identical cells in parallel: total EMF ε, total internal resistance r/n.
- Two cells in parallel: $\varepsilon_{eq} = (\varepsilon_1 r_2 + \varepsilon_2 r_1) / (r_1 + r_2)$, $r_{eq} = r_1 r_2 / (r_1 + r_2)$.

5. Kirchhoff's Rules

Kirchhoff's rules let us analyse circuits that cannot be reduced to simple series and parallel combinations.

Junction rule (current law)

At any junction, the sum of currents entering equals the sum of currents leaving. This follows from conservation of charge: charge cannot pile up at a junction.

Loop rule (voltage law)

Around any closed loop, the algebraic sum of changes in potential is zero. This follows from conservation of energy, because electrostatic potential depends only on position.

Sign convention: moving through a resistor in the direction of current, the potential drops by IR; moving through a cell from its negative to its positive terminal, the potential rises by ε.

Wheatstone bridge

A Wheatstone bridge has four resistors P, Q, R and S arranged in a loop, a cell across one diagonal and a galvanometer across the other. The bridge is balanced when no current flows through the galvanometer, which happens when:

P / Q = R / S

At balance the galvanometer's resistance does not matter. This lets us find an unknown resistance precisely from three known ones.

Meter bridge

A meter bridge is a practical Wheatstone bridge using a uniform 1 m wire. If the balance point is at length l from one end, with known resistance R in the left gap and unknown S in the right gap:

S = R (100 - l) / l (l in cm)

For best accuracy, choose R so that the balance point lies near the middle of the wire, which reduces the effect of end errors.

6. Electrical Energy and Power

When a current I flows through a resistor with potential difference V for a time t, the energy converted to heat is W = V I t. The rate of this conversion is the electrical power:

$$P = V I = I^2 R = V^2 / R$$

The SI unit of power is the watt (W). The heating of a resistor by current is called Joule heating, and it is the principle behind electric heaters, irons and filament bulbs.

Power transmission

Power stations transmit electricity at very high voltages. For a fixed power P = VI, a higher V means a lower current I, and the power lost in the transmission lines, I^2R_c, falls sharply.

Commercial unit

Electricity bills use the kilowatt-hour (kWh), often called a unit. 1 kWh = 3.6×10^6 J. A 1000 W heater running for one hour uses exactly one unit.

7. Quick Revision Table

Quantity	Formula	SI unit
Current	I = q / t	ampere (A)
Current vs drift velocity	I = n e A vd	A
Ohm's law	V = I R	ohm (Ω) for R
Resistance	R = ρ L / A	Ω
Series	Rs = R1 + R2 + ...	Ω
Parallel	1/Rp = 1/R1 + 1/R2 + ...	Ω
Terminal voltage	V = ε - I r	volt (V)
Wheatstone balance	P / Q = R / S	-
Power	P = V I = I²R = V²/R	watt (W)

Common mistakes

- Confusing EMF with terminal voltage: they are equal only when no current flows.
- Thinking electrons move at the speed of the signal: drift velocity is tiny, the field travels fast.
- Assuming every device obeys Ohm's law: diodes and bulbs do not.
- Forgetting that resistivity depends on material and temperature, not on the wire's shape.